DATA CLEANING AND PREPARATION PROJECT

DecodeLabs Data Analytics Internship

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Tool Context: VS Code Environment

Technologies: Python | Pandas | NumPy | Matplotlib

PROJECT OVERVIEW

Objective

The primary goal of this project was to systematically transform a raw, unverified transactional dataset into a pristine, structured, and entirely analysis-ready repository.

Key Project Activities

- **Dataset Inspection:** Initial structural evaluation and orientation.
- **Missing Value Analysis:** Identifying systematic information gaps.
- **Duplicate Record Validation:** Confirming relational integrity.
- **Datatype Verification:** Enforcement of correct data schemas.
- **Outlier Analysis:** Statistical variance assessment via distributions.
- **Final Dataset Preparation:** Production-ready export pipelines.

Project Outcome

A thoroughly validated, high-quality cleaned dataset has been successfully engineered, removing standard analytical risks and preparing the ground for high-fidelity business insights.

DATASET UNDERSTANDING

Dataset Information

The evaluation focuses on a multi-column customer transactions master dataset containing critical fields across several granular dimensions:

- **Order Information:** Transaction identifiers, operational dates, tracking IDs.
- **Product & Pricing:** Product labels, individual quantities, item unit prices, total aggregate calculated prices.
- **Logistics:** Dynamic shipping addresses, specific status logs, selected payment channels.
- **Marketing Metrics:** Coupon codes applied, consumer referral origin metrics.

Initial Structural Assessment

- **Total Rows evaluated:** 1,200 transaction entries.
- **Total Columns monitored:** 14 separate attributes.
- **Initial Quality Status:** Complete schema uniformity, structural anomalies isolated primarily to discount coupon fields.

DATASET PROFILE & SUMMARY STATISTICS

#	Column Name	Non-Null Count	Dtype	#	Column Name	Non-Null Count	Dtype
0	Order ID	1200 non-null	`str`	7	PaymentMethod	1200 non-null	`str`
1	Date	1200 non-null	`datetime64`	8	OrderStatus	1200 non-null	`str`
2	CustomerID	1200 non-null	`str`	9	TrackingNumber	1200 non-null	`str`
3	Product	1200 non-null	`str`	10	ItemsInCart	1200 non-null	`int64`
4	Quantity	1200 non-null	`int64`	11	CouponCode	891 non-null	`str`
5	UnitPrice	1200 non-null	`float64`	12	ReferralSource	1200 non-null	`str`
6	ShippingAddress	1200 non-null	`str`	13	TotalPrice	1200 non-null	`float64`

Statistical Distribution Insights

Metric	Quantity	UnitPrice ($)	ItemsInCart	TotalPrice ($)
Count	1200	1200.00	1200	1200.00
Mean	2.95	356.41	5.49	1053.97
Min	1.00	11.39	1.00	11.39
25%	2.00	186.06	4.00	410.52
50% (Median)	3.00	364.21	5.00	823.62
75%	4.00	521.57	7.00	1578.48
Max	5.00	699.93	10.00	3456.40

DATA CLEANING PROCESS

1. Missing Value Analysis

Executing structural null scanning via `df.isnull().sum()` isolated explicit missing parameters inside the database:

- CouponCode: Identified exactly **309 missing rows**.
- All other 13 features returned absolute zero missing profiles.

2. Missing Value Treatment

Rather than dropping vital financial transactions, missing instances were safely transformed via functional updates:

- Null values replaced globally with descriptive text: "No Coupon".

```
# VS Code Terminal Output Verification
>>> df['CouponCode'].fillna('No Coupon', inplace=True)
>>> df.isnull().sum()
Order ID 0
Date 0
CustomerID 0
Product 0
Quantity 0
UnitPrice 0
ShippingAddress 0
PaymentMethod 0
OrderStatus 0
TrackingNumber 0
ItemsInCart 0
CouponCode 0 (Successfully Handled)
ReferralSource 0
TotalPrice 0
dtype: int64
```

DATA QUALITY & VALIDATION CHECKS

3. Duplicate Analysis

Relational correctness requires evaluating record uniqueness to prevent double-counting transactions.

- Validation function command executed: `df.duplicated().sum()`
- **Result:** Total Duplicate Rows: 0

4. Data Type Verification

Confirmed that formatting alignments perfectly matched expectations across structural types:

- Dates mapped uniformly as `datetime64[us]`.
- Quantities and Cart items verified as `int64`.
- Financial costs and totals kept as precise `float64`.

Key Quality Findings Summary

- Dataset structural borders remained fully intact.
- No missing values remain unhandled.
- Zero row duplication or relational overlap was discovered.
- The baseline data pipeline reflects clean health.

OUTLIER ANALYSIS OVERVIEW

Objective

To systematically evaluate the data for extreme, incorrect, or anomalous statistical deviations that could skew downstream predictive models or data dashboards.

Methods

Constructed boxplot distributions utilizing Matplotlib within our project setup, specifically isolating continuous numerical distributions:

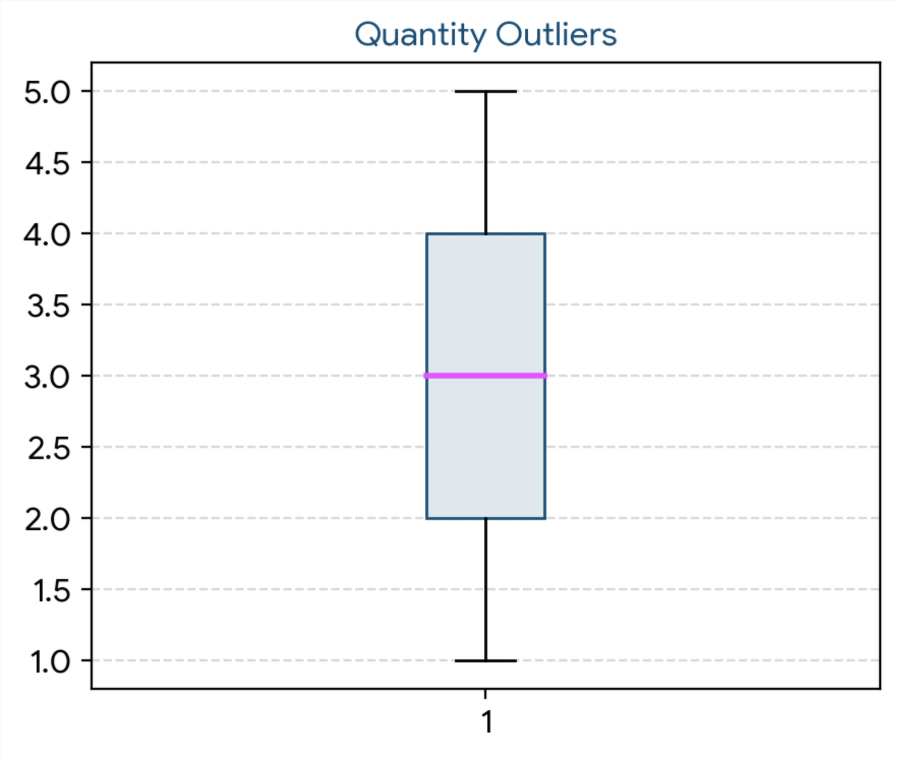
- **Quantity Distributions:** To evaluate purchasing bulk volumes.
- **UnitPrice Distributions:** To review transactional value dispersion.

Visual Observations

Comprehensive inspection of the interquartile range (IQR) and upper/lower fences reveals:

> **No significant outliers were observed.** The values distributed reflect typical operational business ranges, confirming that data clipping or trimming is completely unnecessary.

OUTLIER ANALYSIS VISUALIZATIONS



CONCLUSION

Project Summary

- **Dataset Cleaned:** 1,200 transactional records sanitized.
- **Missing Records Resolved:** 309 missing values addressed in Coupon fields.
- **Zero Redundancy:** Checked and validated for 0 duplicate entries.
- **Data Types Enforced:** Safe database configuration.
- **Anomalies Inspected:** Verified absence of structural outliers.

Final Readiness Status

The engineered dataset is certified accurate and fully optimized for:

`✓ Exploratory Data Analysis (EDA)`

`✓ Interactive Dashboard Development`

`✓ Advanced Predictive Analytics`

Thank You!